

FINTECH

Internship Program

Financial Document Verification System

AI-Assisted Financial Document Processing and Verification Platform

Automating financial document verification using OCR, Computer Vision, Rule-Based Validation, and Vision Language Models while maintaining human approval as the final authority.

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Mentor Review Document

August 4, 2026

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1 Project Overview

1.1 Problem Statement

Financial offices manually verify multiple documents submitted during financial onboarding and payment processing.

The verification process involves checking multiple documents and ensuring that financial information remains consistent across all submitted files.

Documents include:

- Tripartite Agreement
- Bilateral Agreement
- Account Maintenance Certificate
- 1-Link Application Form
- Authority Letter
- Schedule of Charges Agreement
- Business Requirement Document
- Formal Request Letter
- Supporting Documents

The current manual process has several challenges:

- Time consuming verification
- Repetitive checking tasks
- Possibility of human errors
- Difficult cross-document comparison

1.2 Proposed Solution

The proposed system is an AI-assisted financial document verification platform that automates repetitive verification tasks.

The system performs:

- Document uploading
- Document completeness checking
- Technical file validation
- Text extraction
- Information extraction

- Business rule validation
- Signature and stamp detection
- Validation report generation

The employee remains the final authority.

The employee can:

- Approve
- Correct
- Reject

2 Business Rules

2.1 Required Document Verification

The system verifies the presence of:

- Tripartite Agreement
- Bilateral Agreement
- Account Maintenance Certificate
- 1-Link Application Form
- Authority Letter
- Schedule of Charges Agreement
- Business Requirement Document
- Formal Request Letter

2.2 IBAN and Account Consistency

The same bank account information must exist in:

- Account Maintenance Certificate
- Bilateral Agreement
- Tripartite Agreement

The system checks:

- IBAN format correctness
- IBAN matching across documents
- Account number format
- Bank information consistency

2.3 Signature Verification

The system verifies the presence of signatures in required locations.

Required documents:

- Authority Letter
- Account Maintenance Certificate
- 1-Link Application Form
- Tripartite Agreement
- Schedule of Charges Agreement
- Business Requirement Document
- Formal Request Letter

2.4 Stamp Verification

The system verifies official stamp presence in:

- Account Maintenance Certificate
- 1-Link Application Form
- Tripartite Agreement
- Schedule of Charges Agreement

2.5 Field Validation

The system verifies:

- Required fields are filled
- Correct data formats
- CNIC format
- CNIC expiry
- Account number format
- Date formats

2.6 Human Verification

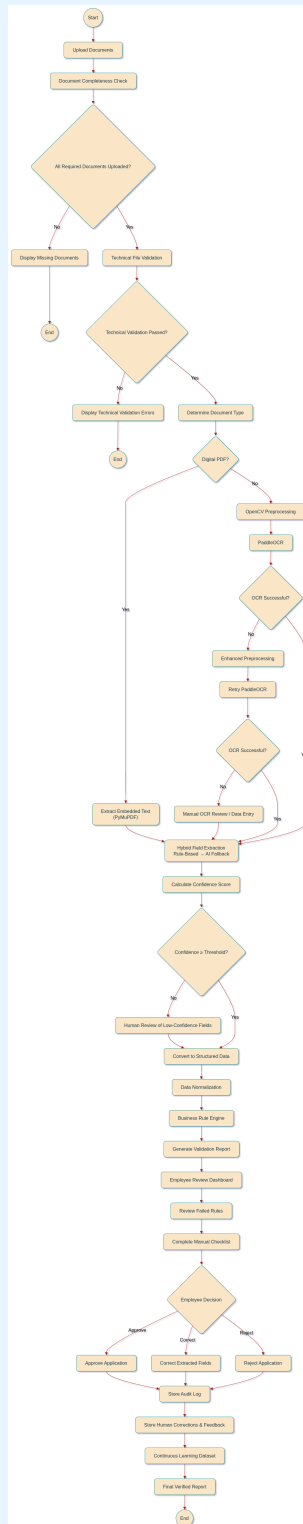
The employee verifies:

- Originality
- Visible tampering
- Signature detection results
- Stamp detection results
- Low confidence fields

Final decisions:

- Approve
- Correct
- Reject

3 System Pipeline



Complete AI-Assisted Financial Document Verification Pipeline

4 Technology Stack

The system follows a hybrid AI architecture combining deterministic validation with artificial intelligence.

Rule-based methods handle structured financial checks, while AI models are used as fallback mechanisms for complex visual and language understanding tasks.

The architecture follows a human-in-the-loop approach where the employee remains the final decision authority.

Component	Technology
Frontend	React.js / Streamlit
Backend API	FastAPI
Programming Language	Python 3.12+
Database	PostgreSQL
PDF Processing	PyMuPDF
Image Processing	OpenCV
OCR Engine	PaddleOCR
Field Extraction	Regex + Document Templates
AI Fallback	Qwen2.5-VL
Signature and Stamp Detection	YOLO / Qwen2.5-VL
Validation	Pydantic
Business Logic	Custom Python Rule Engine
Logging	Python Logging
Version Control	Git/GitHub

5 Development Phases

Phase 1: Project Foundation
Architecture setup, FastAPI configuration, database connection, environment configuration and logging.

Phase 2: Database Architecture
Design storage for applications, documents, extracted data, validation results, feed-back and audit logs.

Phase 3: File Upload System

Secure document upload, file validation and metadata storage.

Phase 4-5: Document Verification Foundation

Completeness checking, technical file validation, PDF/image quality analysis and readability checks.

Phase 6-8: AI Processing Pipeline

Digital PDF extraction, OCR processing, preprocessing, hybrid information extraction and confidence scoring.

Phase 9-10: Validation Intelligence

Data normalization, business rule engine, cross-document verification, signature and stamp detection.

Phase 11-12: Human Verification

Validation reports, employee review interface, correction and approval workflow.

Phase 13-14: Feedback and Continuous Improvement

Store verified corrections and build future improvement datasets without automatic retraining.